



Strategic emerging industries and innovation: Evidence from China[☆]

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ABSTRACT

This paper relies on the policy of strategic emerging industries in China as a quasi-natural experiment to examine the role that the industry policy plays in shaping firm innovation. Utilizing a listed firm-level panel dataset over 2006–2020 in difference-in-differences framework, we find that innovation increases after this policy implemented, and this increase is larger in firms belonging to the strategic emerging industries. Our empirical results hold across various econometric specifications employed and support the argument that the industry policy strengthens the incentives of firms to innovate. Moreover, we find evidence that the strategic emerging industries policy significantly promotes tax benefits, firm loans, industrial concentration, R&D input and innovator quality, leading to more innovative outcomes. This paper also suggests that developing countries closed to the frontier of world's advanced technologies can adopt the industry policy to optimize the innovative outcomes structure.

1. Introduction

From 1980 onwards, China's economy grew at a spectacular annual rate based on several key factors: the imitation of technology invented but obsoleted by developed countries and a sequence of market-oriented institutional reforms, combined with low wages and a decent demographic structure.

However, due to the problems of rising wages, environmental pollution, extensive production methods as well as approaching to the frontier of developed countries' technology (i.e., the United States and European countries), the mode of economic development based on imitation and the input of factors is hardly sustainable. As a result, the State Council of China released the *Outline of the National Medium-to-Long Term Science and Technology Development Program (2006–2020)* in 2006, and the *Outline of the National Intellectual Property Strategy* in 2008 with the aim for China to become an 'innovation-oriented' country by 2020 and a country with 'leading science power' by 2050. The purpose of the outline is to improve the ability of China to create, utilize, protect and administer intellectual property. This suggests that China's economy steps into a new era of quality-concentrated development and innovation.¹ The moderation of growth since 2012 leaves a hint that China's economy transforms from quantity-focused into quality-oriented

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¹ Top leaders in China treat innovation as one key driving force for the country's future economic growth. For example, innovation is listed as the first guiding principle of economic policy in the 13th Five-Year Plan released in 2016.

development. Thus, a decisive factor for China's future economic development is whether its firms can innovate and upgrade quality and the pace they proceed at.

It is widely known that innovation plays a crucial role in economic growth (Romer, 1990). Nonetheless, due to the nature of public goods in innovation, companies are short of inspiration and motivation to conduct innovative activities (Nelson, 1959). In addition, the uncertainty, non-appropriability and non-divisibility of the innovation outcomes also deter companies to fully internalize their returns on investment. As a result, industry policy, as one tool designed by the Chinese central government in the indigenous innovation based way, is utilized to assure latecomers to economically and technologically catch up with developed countries (Gerschenkron, 1962). Given that innovation is the essential tool to push catch-up as in the neo-Schumpeterian view (Nelson, 1985), state-led innovation is still more suitable for China to drive catch-up relative to the uncoordinated purely market-responsive method (Liu et al., 2011).

The most significant recent industrial policy in China is the “strategic emerging industries” initiative. The strategic emerging industries outlined by the central government consist of a group of seven industries: energy conservation and environmental protection, information, biology, high-end equipment manufacturing, new energy, new materials, and new energy automobiles. This development is a crucial part of China's state-led catch-up strategy after the 2008 global financial crisis. The demand of domestic and global market is envisaged to be high in these strategic emerging industries by 2020. In addition, the development of these industries is intended to solve the challenges in socio-economic and environmental areas in China.

The strategic emerging industries initiative was proposed during a number of seminars in 2009 organized by the State Council Premier and Vice Premier. This was then officially issued by the State Council in the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries* on October 10th 2010. This decision sets targets for strategic emerging industries to increase their contribution to 2% of GDP to 8% by 2015, and 15% by 2020. To achieve this target, the Chinese government had planned to spend 10 trillion RMB over the period 2011 to 2015 to develop strategic emerging industries. The *State Council Notice on Printing and Distributing the 13th Five-Year National Strategic Emerging Industries Development Plan* emphasizes the importance of supply-side innovation. As the core driver of strategic emerging industries, innovation is critical to their development. The plan advocates for a comprehensive implementation of an innovation-driven development strategy, which includes increasing the supply of technology, talent, and capital. It aims to create an ecosystem that fosters interaction and integration of innovation factors, to break through key core technologies, enhance independent innovation capabilities, reduce enterprise costs, stimulate enterprise vitality, and accelerate the growth and strengthening of emerging enterprises.

This paper focuses on a panel of data covering 4,161 Chinese listed firms over the period 2006–2020. The central empirical question we examine is how the strategic emerging industries initiative affects Chinese firms' innovative output (i.e. the number of granted patents). This paper adopts a difference-in-differences method to analyze the impact of strategic emerging industries initiative on innovation. Specifically, we compare the innovative output of listed firms belonging to the group of seven industries in the strategic emerging industries, with that of listed firms belonged to the industry without this title. Due to the fact that the process of selection of these industries is supervised by the central government, this paper utilizes this industrial policy and treats it as a quasi-natural experiment to test the effect of the strategic emerging industries initiative on firms' innovation.² This paper also scrapes the data for IPR related trials through the China Judgement Online to examine how this industrial policy affects innovation through the mechanism of judicial environment.

Our analysis makes use of listed firms and manual collection data and adopts the methods of alternative measures of dependent variable, propensity score match, and dynamic effects as well as a series of robustness check to document three main findings. First, the increase in innovation is significantly larger in firms belonging to the strategic emerging industries than in firms outside this group of industries. On average, the number of total granted patents rises by 27.8% in the strategic emerging industries compared to the counterpart. Second, we find evidence that this industrial policy significantly improves the level of tax benefits, firm loans, industrial concentration, R&D investment and innovator quality, leading to more innovative outcomes. The impact of strategic emerging industries is induced by the State Council development strategy that motivates research and development investment by firms. Third, the rise in innovation is greater in cities with higher degree of IPR protection, and innovation. In addition, this industrial policy plays a key role in the structure optimization effect on earlier firms or non-state-owned enterprises. In contrast, the number of total granted patents in later stage firms or state-owned enterprises increases together with an adjustment of the innovative outcomes structure. The scientific background of executives is confirmed to have an effect on firm innovation.

We believe this paper makes two crucial contributions to the existing literature. First, our analysis attempts to examine the effect of the strategic emerging industries policy on firm innovative outcomes in China. In an era of economic transition in China, the price of the factors of production shows an increased trend. The strategic emerging industries policy plays a significant role in China's economic transition. Thus, this paper leaves a hint on the debate that whether high or low quality of industrial policy is appropriate for developing countries. Second, we present the policy of strategic emerging industries as one new explanation of the innovation miracle in China, given its rapid innovative development in recent years.³ It is widely regarded that China innovates through means of imitation, whilst from the perspective of this paper the industrial policy still plays a key role in innovation outcomes in China. Additionally, this industrial policy has a structure optimization influence, which also sheds light on developing countries which aim at improving their innovation quality.

This paper is also related to the literature on the policy of strategic emerging industries in China (Zhao et al., 2019). Zhao et al. (2019) use the same policy change in 2010 (the designation of certain sectors as strategic emerging industries) to study the same

² Fig. 1 displays the trend between treatment group and control group used in the analysis.

³ Wei et al. (2017) fully discuss the miracle of innovation in China — from “Made in China” to “Innovated in China”.

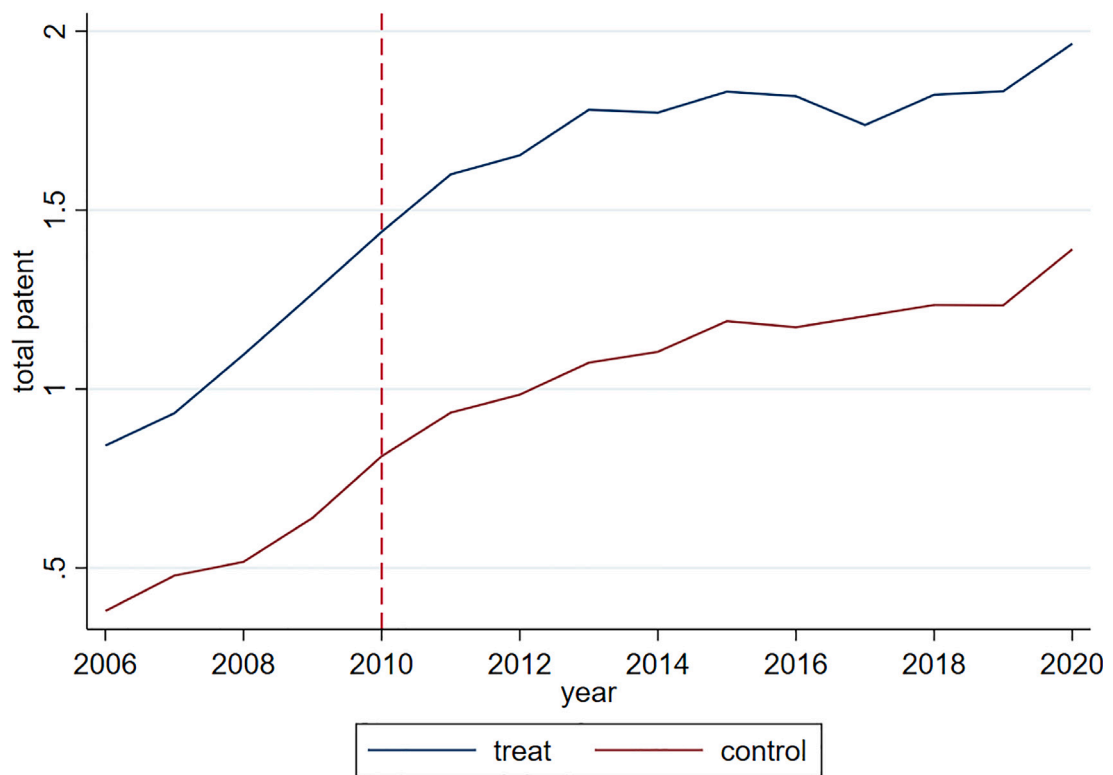


Fig. 1. Trend between treatment group and control group.

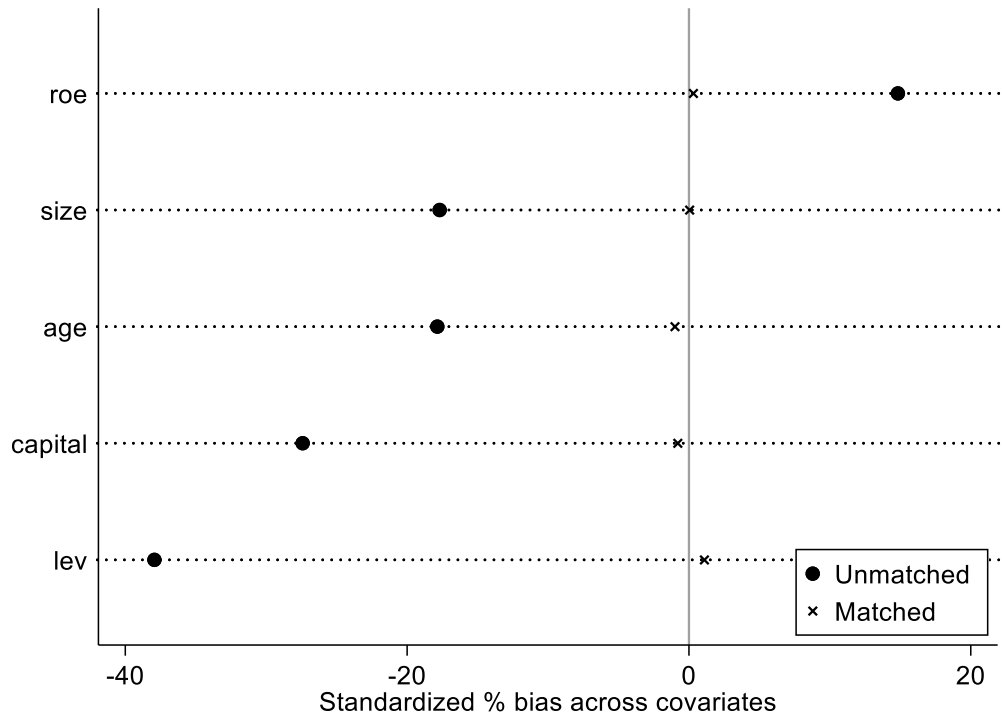


Fig. 2. Propensity score matching.

outcome variable (the number of patents, in particular invention patents) using the same set of methods. The policy of strategic emerging industries we investigate on differs from their assessment in at least two ways. First, the main noteworthy difference is that Zhao et al. (2019) focus on the new energy vehicle industry as the treated group, while our analysis uses all strategic emerging industries as the treated group. Second, we further examine the structure optimization influence after the implementation of strategic emerging industries policy by utilizing different categories of patents.

The next section introduces the existing literature. Section 3 describes the data and the methodology. Section 4 contains the estimation results, and Section 5 concludes.

2. Literature review

2.1. Strategic emerging industries

Strategic emerging industries, as the key industries to be cultivated in China at the current stage, are those industries that are established based upon major breakthroughs obtained advanced science and technology. These industries also represent the novel direction of future science and technology development, reflect the trend of knowledge, low-carbon, and circular economy development, have great potentials even in the early stage, and play a key role in leading economy and society. As strategic emerging industries are crucial forces to lead the future development in society and economy, the development of these industries have become a key strategy of major economies to catch the opportunity of the new round of development. To achieve the target proposed in China's 13th Five-year Plan and improve the endogenous power of the strategic emerging industries development, it is necessary to improve the innovation ability.

Since research and development investment is the fundamental method to enhance the innovation level of strategic emerging industries, it is significant to transform the economic development pattern and improve industrial competitiveness (Xu et al., 2020). Due to a great deal of capital with high risk and long period of return needed for the research and development investment, firms face difficulty in financing, such as coming from internal fund and external financing.⁴ The higher cost of external financing compared to that of internal financing results in financing constraints on research and development investment. However, the existing literature working on the financing constraints faced by firm research and development investment has yet come to a consensus. Fazzari et al. (1988) explore the effect of financing constraint on firm investment given the assumption of imperfect capital market and asymmetric information. Later on, a series of literature finds a positive and significant correlation between firm research and development investment and internal cash flow (Hu et al., 2017; Sasidharan et al., 2015), whilst others conclude that there is a significant negative relationship between firm research and development investment and financing constraints (Chen & Yang, 2018; Zhang, 2018). Instead, Liu et al. (2018) find a nonlinear relationship between these two factors. If the level of financing constraints is high, financing constraints are positively correlated with firm research and development investment, and if the level of financing constraints declines to a certain threshold, the constraints become negatively related to research and development investment.

In addition, current studies working on the strategy of development for latecomer economies prefer using industrial policy aimed at specific industries depending on economic and technological factors.⁵ Lin (2012) argues that poor economies are supposed to specialize in industries reflective of their initial endowments such as labor and natural resources or comparative advantages related to resource endowments. Greenwald and Stiglitz (2013) favor specialization in industries with relatively higher level of endowments of local knowledge and a propensity to localize knowledge in a quicker way. Further studies then extend these concepts into the argument of capability failure for government intervention according to the experiences of South Korea, Taiwan, and other latecomers (Lee & Mathews, 2010, 2012). As for specialization strategies, latecomer firms and countries should have an adequate base degree of technological capability and experience before attempting leapfrogging particularly, or they will be difficult to conduct a sustainable catch-up in the long-term (Sohn et al., 2009).

The decision to choose to specialize strategic emerging industries also depends on the economic status of the region that firms are located in. Economies with smaller size and higher level of specialization start with lower degree of income per capita and research and development intensity, growing at higher rates compared with other latecomers (Pianta & Melicani, 1996). Due to lower level of technological base, small economies are likely to embrace higher level of technological specialization, and confront with the difficult but necessary mission to adjust their specialization policy when facing the structural technological change (Soete, 1988). Moreover, smaller economies are to some extent compelled to specialize in certain niches and consequently, they tend to be more dependent on international technological linkages (Archibugi & Pianta, 1996).

2.2. Innovation quality

A series of literature focuses on the effect of industry policy on innovation quality, yet the research findings remain inconsistent. Some scholars argue that industrial policy exerts a significant positive effect on innovation. For instance, Garcia-Macia et al. (2019) investigate the effects of innovation subsidies as a tool of industrial policy on corporate R&D activities. They find that innovation subsidies can effectively address the issue of under-investment in cutting-edge technology research, leading to substantial

⁴ A number of recent studies (i.e., Feng et al., 2023; Luo et al., 2022; Luo & Zhang, 2022; Qin et al., 2022) further concentrate the influence of Chinas local political turnover or policy uncertainty on firm investment. However, Zhang et al. (2021) instead pay attention to the importance of local political leaders during the prevention and control period of COVID-19 in China, given that evidence abounds of the local political turnover in China (Luo & Qin, 2021).

⁵ Shen et al. (2021) also work on the heterogeneous effect of location characteristics when firms face a shock of environmental regulation.

improvements in corporate innovation capacity and productivity. Similarly, Feng (2019) analyzes a sample of A-share listed companies in Shanghai and Shenzhen from 2011 to 2015, examining how industrial policy and labor allocation impact corporate innovation efficiency. His results indicate that industrial policy plays a crucial role in enhancing innovation efficiency, particularly in firms with inadequate human resource allocation.

Conversely, other scholars contend that the impact of industrial policy on innovation is either limited or even counterproductive, with outcomes varying depending on the type of policy. Noland and Pack (2003), for example, study industrial policy in Japan and Korea and find no evidence that specific R&D subsidy activities significantly affect national or industry productivity. Catozzella and Vivarelli (2011) explore the effects of government subsidies on research and development inputs and outputs among Italian firms. Their findings suggest that while government subsidies significantly boost R&D inputs, but they have an insignificant impact on innovation outcomes. Additionally, Falck et al. (2010) investigate the effects of cluster-oriented industrial policies in Bavaria, Germany, and discovered that while these policies can enhance regional enterprises' innovation capacity, they may also reduce firms' independent R&D investments.

The reasons behind these differing results may be related to the type of industrial policy implemented. Wang and Zou (2018) analyze the impact of various types of industrial policies on corporate innovation and find that demand-side and environment-side policies tend to inhibit innovation capacity, whereas supply-side policies have a significantly positive effect, especially on core technology innovation. Consequently, they argue that policy-making should place greater emphasis on supply-side support.

China's inventions are of generally low value and quality (Huang et al., 2021; The Economist, 2010, 2013). However, looking behind the headline numbers raises doubts about the quality of China's patents and the degree to which the surge reflects true innovation. For example, the vast majority of Chinese patents do not relate to actual inventions, but instead to designs and minor improvements (Liang, 2011). Many patents in China – particularly those filed by domestic applicants – appear not to be transformed into something useful, or at least could be transformed into something more useful. As such, they do not meet or could better meet the second criterion in the definition of quality (and highest-quality) patents. Patents in force and lifespans of patents are some of the best available metrics to assess this trend (Prud'homme, 2012a).

The results in Thoma (2013) provide evidence supporting the “strategic patenting” hypothesis on the lower value and quality of Chinese patents compared to other patents on average, though a few caveats are in order. Both domestic and foreign firms file patents not only protect real products in the market but also for strategic reasons (Hu, 2014).

The analysis in Prud'homme (2012b) suggests there are weaknesses in certain targets due to the absence of important criteria for ensuring patent quality. Further, the overly heavy focus on just a few types of quantitative patent targets (e.g., for patent applications and patents granted) overshadows the type of benchmarking that better reflects conditions stimulating creativity as well as the actual economic, social, and environmental relevance of the underlying inventions (i.e., their ability to be transformed into something useful and thus constitute innovation).

With the policy of strategic emerging industries being implemented by the State Council of China, this paper aims to quantify the impacts of industrial policy on innovative outcomes from a more profound perspective of micro-economic effects to fill the existing research gaps.

2.3. Theoretical framework

In the Resource Dependence Theory (Pfeffer & Salancik, 1978), Casciaro and Piskorski (2005) argue that the “interdependence” between organizations can be divided into two parts: “co-dependence” and “power imbalance”. For state-owned enterprises, on the one hand, there is a clear “mutual dependence” relationship between state-owned enterprises and the government. State-owned enterprises rely on the government to obtain policy incentives and support, while the government also relies on state-owned enterprises to achieve goals such as taxation, employment, and social security (Liao & Shen, 2014). On the other hand, the “power imbalance” between state-owned enterprises and the government is also evident, which means that the strong side will have a significant impact on the weak side (Casciaro & Piskorski, 2005). In state-owned enterprises, state-owned capital plays an absolute dominant role, which means that most of the key resources needed for the survival and development of the enterprise come from state-owned capital, and state-owned enterprises are highly dependent on state-owned capital in terms of resources. Therefore, in the process of corporate investment, state-owned enterprises are likely to exhibit two contradictory behavior patterns of “eager performance” and “inaction” simultaneously (Jin et al., 2016). As a disadvantaged party in resource dependence, state-owned enterprises will become active responders to government policies (Liu, 2018). If state-owned enterprises comply with the government's internationalization strategy in foreign investment, they will receive policy support from the government (Cui & Jiang, 2012).

Luo (2025) elaborate the cooperation between government and social capital from the perspective of reverse mixed ownership reform. Mixed ownership is one of the methods to reduce the dependence of state-owned enterprises on government resources (Choudhury & Khanna, 2018). Due to mixed ownership, enterprises are composed of both state-owned and non-state-owned capital. Under the control of state-owned capital, enterprises tend to associate their business goals with government objectives such as economic development and political goals (Liu, 2018), while non-state-owned capital places more emphasis on the pursuit of operating profits, market share, or resources (Rodrigues & Dieleman, 2018). This means that not all key resources for the survival and development of enterprises come from state-owned capital or government departments, and non-state-owned capital plays a balancing role on state-owned capital. Therefore, the existence of state-owned capital within mixed ownership enterprises will to some extent have a call to comply with government policies and accelerate investment; At the same time, the existence of non-state-owned capital with profit driven characteristics gives enterprises a strong motivation to pay attention to operational risks and development dynamics, enhancing their market competitiveness (Ma et al., 2015).

Due to a great deal of capital with high risk and long period of return needed for the research and development investment, firms face difficulty in financing. When a firm cannot insure against uncertainty, it has incentives to reduce its capital stock to avoid the probability of negative payout for a project. Luo et al. (2022) reckon this to be a typical risk-avoiding action. According to the Resource Dependence Theory, tax benefits and credit support can to some extent alleviate financing constraints and reduce the risks tied to innovation.

3. Data and methodology

3.1. Data and variable definition

This paper focuses on the data of listed companies in China A-shares market over the period 2006–2020. We compile the dataset derived from three different sources. First, due to the lack of classification regarding strategic emerging industries in the China Securities Regulatory Commission, the definition in our analysis refers to the Emerging Composite Index (*xinxing zongzhi*) released by the Shanghai Stock Exchange⁶ and the China Securities Index Co., Ltd.⁷. Specifically, the strategic emerging industries outlined by the index consist of a group of industries: energy conservation and environmental protection, new generation information technology, biology, high-end equipment manufacturing, new energy, new materials, new energy automobiles, digital creative, and high-tech service. Compared with other studies in the literature, this definition of strategic emerging industries adheres to the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industry*.⁸ Therefore, firms classified as strategic emerging industries in the following analysis are extracted from this index. Second, the data for listed firms' granted patents in different categories and financial characteristics are taken from the China Stock Market and Accounting Research (CSMAR) database. Third, we scrape the IPR related trials data from the China Judgement Online,⁹ which was initiated by the Supreme People's Court in 2013 and then extended to contain all judgement documents published by China's four-level courts in 2014. Since all valid judgement documents in China were uploaded to this website, we construct the web spider via python program to collect the number of IPR related trials over 2006–2020 according to the cause of action, and treat the location of the court handled as the geographic distribution of the trials. In sum, after the exclusion of missing values, we finally obtain 36,079 observations, covering 4,161 listed companies. There are 1,212 firms affected by the industry policy, accounting for 29% of total sample.

This paper aims to examine how the policy of strategic emerging industries affects innovation in China. As for the dependent variable, we use the number of granted patents to measure firm-level innovation. Due to the non-negative values and skewed distribution, we use the natural logarithm of (one plus) the number of granted patents to alleviate its skewness, following Zhang et al. (2023). According to the category, this paper also uses the following types of patents as alternative measurements of innovation in the econometric analysis below. The first patent category is invention patents, which occupy the highest innovative content since they incorporate novel technologies. The second category is utility patents, accounting for new applications of existing technologies. The last one is design patents, presenting novel packaging and design. Among these three categories, invention patents are widely regarded as “high quality”, with relatively higher level of innovative value and practical uses. According to Tong et al. (2014) and Hall and Harhoff (2012), listed firms are more likely to pursue “quantity” rather than “quality” owing to the incentive of rent-seeking, leading to a behavior of strategic innovation. Therefore, this paper utilizes different category of patents to investigate the existence of strategic innovation behavior during the implementation of industry policy.

This paper uses the policy of the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries* issued by the State Council of China in 2010 as a quasi-natural experiment and adopts a difference-in-differences approach to analyze the effect of the strategic emerging industries policy on firm innovation. Therefore, the key explanatory variable in this paper is the interaction term of a dummy whether the firm belongs to the category of strategic emerging industries and a time dummy whether it is after the implementation of the policy, 2010.¹⁰

In addition, in the following regression analysis, we use a number of firm-level control variables that have effects on firm innovation. First, the natural logarithm of (one plus) the number of firm operating years (Age) is employed to measure firm age. Second, we use the number of employees (Size) to measure the size of the firm. Third, return on equity (ROE) is employed as a control variable to reflect firm operation condition. Fourth, firm asset-liability ratio (Leverage) is included to measure firm risk level. Fifth, firm fixed asset ratio (Capital) is added as one control to reflect firm capital stock.

In terms of mechanism, due to the uncertainty and high-risk of innovative outcomes, appropriate government support can to some extent reduce the cost of firm research and development, encouraging their input of research and development (Greenwald &

⁶ <http://english.sse.com.cn> (accessed July 13, 2021).

⁷ <http://www.csindex.com.cn/en> (accessed July 13, 2021).

⁸ The definition of strategic emerging industries in the existing literature has at least two concerns. First, one may use the definition in the Wind database, with only 497 firms classified as strategic emerging industries. However, there are more than 1100 firms according to the official release. According to our definition, there are 1,212 firms classified as strategic emerging industries, which highly coincides with the official release. Second, other studies may utilizing the expert scoring method. However, compared with the Emerging Composite Index, this method could be highly subjective.

⁹ <http://wenshu.court.gov.cn> (accessed July 13, 2021).

¹⁰ The paper assumes that the strategic emerging industries policy impacted all relevant sectors uniformly from 2010 onward. Admittedly, policy implementation might likely vary across industries. Addressing this variability would strengthen the robustness of the empirical findings. While related information is unavailable within the policy of the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries*, we have to adopt the “one-size-fits-all” assumption, and acknowledge that our results may overlook critical policy effectiveness nuances. Zhao et al. (2019) also utilize the same assumption when looking at the strategic emerging industries policy.

Table 1
Descriptive statistics.

	Definition	Obs.	Mean	Std.dev.	Min	Max
ln(granted patent+1)	log of (one plus) the number of granted patent	38,645	1.232	1.461	0	8.906
ln(granted invention patent+1)	log of (one plus) the number of granted invention patent	38,645	0.578	0.966	0	8.034
ln(granted utility patent+1)	log of (one plus) the number of granted utility patent	38,645	0.858	1.272	0	8.349
ln(granted design patent+1)	log of (one plus) the number of granted design patent	38,645	0.343	0.888	0	6.925
ln(applying patent+1)	log of (one plus) the number of existing patent	38,645	1.388	1.600	0	9.358
ln(applying invention patent+1)	log of (one plus) the number of existing invention patent	38,645	0.933	1.285	0	8.784
ln(applying utility patent+1)	log of (one plus) the number of existing utility patent	38,645	0.882	1.300	0	8.382
ln(applying design patent+1)	log of (one plus) the number of existing design patent	38,645	0.342	0.892	0	6.815
ln(tax refunds+1)	log of (one plus) tax refunds	33,199	12.808	6.691	0	23.381
ln(total loan+1)	log of (one plus) firm short loan	29,631	17.938	6.644	0	26.749
Lerner	industry Lerner index	32,675	0.118	0.077	−0.224	1
R&D intensity	ratio of R&D expense to profit	38,645	0.121	0.389	−0.322	2.572
Graduate ratio	ratio of the number of employees with graduate degree to total employees	19,217	0.055	0.058	0	0.240
ln(average wage+1)	log of (one plus) average wage	35,469	11.347	0.582	5.774	12.420
Age	the number of firm operating years	38,645	16.968	6.241	0	70
Size	log of (one plus) total asset	37,858	22.027	1.328	19.135	27.059
ROE	return on equity	37,812	0.035	0.061	−0.380	0.206
Leverage	asset–liability	37,809	0.448	0.217	0.051	1.136
Capital	fixed asset ratio	37,849	0.216	0.163	0.001	0.724

Notes: The table gives descriptive statistics for the variables.

Table 2
Basic estimation results.

DEP VAR=log of	(1) granted patent+1	(2) granted invention patent+1	(3) granted utility patent+1	(4) granted design patent+1	(5) granted patent+1	(6) granted invention patent+1	(7) granted utility patent+1	(8) granted design patent+1
Teat × Post	0.280** (0.116)	0.335*** (0.091)	0.156* (0.088)	0.070* (0.040)	0.245** (0.115)	0.316*** (0.091)	0.125 (0.088)	0.060 (0.041)
ROE					−0.235** (0.090)	−0.283*** (0.084)	0.008 (0.082)	−0.006 (0.080)
Size					0.070** (0.031)	0.061** (0.025)	0.053** (0.025)	0.031** (0.014)
Leverage					0.066 (0.071)	0.075 (0.049)	0.048 (0.053)	0.031 (0.041)
Capital					0.369*** (0.109)	0.292*** (0.082)	0.237*** (0.087)	0.127*** (0.044)
Age					0.033*** (0.006)	0.016*** (0.004)	0.029*** (0.005)	0.006*** (0.002)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	38,645	38,645	38,645	38,645	36,079	36,079	36,079	36,079
R-squared	0.047	0.073	0.033	0.006	0.053	0.081	0.037	0.007
Number of firms	4,230	4,230	4,230	4,230	4,161	4,161	4,161	4,161

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

Stiglitz, 2006; Lichtenberg & Siegel, 1991). Firm innovative outcomes rise with this input, indicating the development of innovation. Therefore, this paper analyzes the cause of the growth of firm innovation output in the strategic emerging industries by means of tax refunds, and loan, to reveal the channels that the industry policy affects firm innovative activities. Table 1 reports the definition and summary statistics of all the variables. As we can see, the standard deviation of all categories of patents is higher than their average value, demonstrating that there are quite different in the number of granted patents among firms.

Table 3

Robustness — propensity score matching.

DEP VAR=log of	(1) granted patent+1	(2) granted invention patent+1	(3) granted utility patent+1	(4) granted design patent+1
Treat × Post	0.253** (0.119)	0.314*** (0.092)	0.135 (0.092)	0.062 (0.041)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes
Method	PSM	PSM	PSM	PSM
Observations	35,374	35,374	35,374	35,374
R-squared	0.053	0.080	0.038	0.007
Number of firms	4,154	4,154	4,154	4,154

Notes: Estimations construct propensity score matching (PSM) with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

3.2. Econometric specification

The basic framework of our analysis is to explore the casual relationship between the policy of strategic emerging industries and innovation at the firm level. To sum up, the model central to this is

$$\ln(\text{granted patent} + 1)_{i,t} = \beta_0 + \beta_1 \text{Treat}_i \times \text{Post}_t + \mathbf{x}'_{i,t} \Gamma + \alpha_i + \eta_t + u_{i,t} \quad (1)$$

where i represents each firm, t represents each time period, and $u_{i,t}$ is the error term. The left-hand-side variable, $\ln(\text{granted patent} + 1)$, is a measure of innovation at firm i in year t . In addition to the key dependent variable, log of (one plus) the number of total granted patent, alternative dependent variables (i.e., different categories of patent as described above) are also employed in this paper to examine this relationship. Treat_i is a dummy variable that equals one if the firm i belongs to the strategic emerging industries outlined by the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries*, and zero otherwise. Post_t is a dummy variable that equals one if time t is after the policy implementation of strategic emerging industries, 2010, and zero otherwise. $\text{Treat}_i \times \text{Post}_t$ is the interaction term of Treat_i and Post_t , and the coefficient β_1 captures the average treatment effect of industry policy on firm innovation. A positive and significant β_1 suggests that the policy of strategic emerging industries exerts a positive effect on innovative outcomes, whilst a negative and significant β_1 indicates that the industry policy pushes the level of innovation lower. Control variables analyzed above (such as firm-level basic and financial information) are included in the vector $\mathbf{x}_{i,t}$.

We also include year-specific dummy variables, η_t , to control for shocks and trends that shape innovation over time, such as business cycles. Then we also include firm-specific dummy variables, α_i , to control for time-invariant, unobserved firm characteristics that shape innovation across firms. Stand errors are clustered at the industry level due to the potential correlation of error term, u , within an industry.

The methodology used above quantifies the average treatment effect of industry policy, but it can hardly trace out the trends after its implementation. As a result, we use the following identification strategy to conduct the dynamic effects:

$$\ln(\text{granted patent} + 1)_{i,t} = \beta_0 + \sum_{k=-3}^4 \beta_k \text{Treat}_i \times \tau_k + \mathbf{x}'_{i,t} \Gamma + \alpha_i + \eta_t + u_{i,t} \quad (2)$$

where k represents the time interval of the implementation of the policy of strategic emerging industries. τ_k is a dummy variable taking the value of one if the time interval equals k ; otherwise, it is zero. After detrending and centering the estimates one period prior to the policy implementation, we can capture the results for dynamic effects. Thus, the key coefficient β_k represents the average treatment effects in different time periods.

4. Empirical results

4.1. Baseline regressions

This section is to test whether and how innovation across different firms systematically changes with the policy of strategic emerging industries. Column 1 of Table 2 is a simple specification with a measure of industry policy ($\text{TREAT} \times \text{POST}$), in the presence of fixed firm and year effects, using annual data regression, with robust standard errors clustered by industry. This shows a positive and significant relationship between the industry policy and total granted patents. This is consistent with our argument — an improvement in the cultivation and development of strategic emerging industries promotes firm innovation. As argued above, column 2 then extends column 1 to use the number of granted invention patents as an alternative dependent variable. In this specification the sign of the coefficient estimate relating to the industry policy is positive, and statistically significant at the 1% level. Columns 3 and 4 further utilize the category of patent and respectively employ utility and design patents as alternative measures of innovation, whilst the estimated statistical significance of industry policy declines. Columns 5–8 then in sequence extend columns 1–4 to include a set of firm-level characteristics as control variables. The results using full controls support those already found.

Table 4
Dynamic effects.

DEP VAR=log of	(1) granted invention patent+1	(2) granted invention patent+1	(3) granted invention patent+1
Treat_tb3	−0.059 (0.052)	−0.058 (0.053)	−0.049 (0.051)
Treat_tb2	0.029 (0.032)	0.046 (0.029)	0.048 (0.029)
Treat_t0	0.128*** (0.041)	0.130*** (0.038)	0.132*** (0.039)
Treat_ta1	0.217*** (0.058)	0.225*** (0.064)	0.217*** (0.060)
Treat_ta2	0.275*** (0.066)	0.269*** (0.070)	0.264*** (0.068)
Treat_ta3	0.264*** (0.077)	0.256*** (0.079)	0.261*** (0.075)
Treat_ta4	0.390*** (0.101)	0.370*** (0.100)	0.371*** (0.094)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Control Variables	No	Yes	Yes
Provincial Fixed effects × Year Fixed effects	No	No	Yes
Observations	38,645	36,079	36,079
R-squared	0.077	0.084	0.105
Number of firms	4,230	4,161	4,161

Notes: Estimations test dynamic effects with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

Using the estimate from the second half of [Table 2](#), the number of total granted patents on average rises by 27.8% in the strategic emerging industries compared to the counterpart, holding all else equal. Among three categories of patent, the rise in the number of granted invention patents is the most obvious, accounting for 27.6%, holding all else equal. This is followed by the utility patents and the design patents. Both the change in the number of granted utility and design patents are insignificant. The results indicate that the implementation of the strategic emerging industries policy has a positive effect on firm innovative outcomes. This leads to more research and development activities by firms belonging to the strategic emerging industries, achieving that a rapid growth in the number of granted invention patents.¹¹

4.2. Robustness tests

[Table 3](#) contains the robustness test. Since the method of difference-in-differences uses the control group as the contra-factual of the treatment group to conduct the identification of causal relationship, the characteristic variables in the treatment group and the control group are not supposed to have significant differences in order to assure the reliability of identification. Therefore, we utilize the method of k-neighbor matching to adopt propensity score matching, with an exclusion of the sample which is not in the common area, and then conduct the difference-in-differences regression. To be specific, utilizing the firm-level characteristic variables (*i.e.*, ROE, Size, SOE, Cash, Leverage and Age) in the baseline estimation, we use the logit model to estimate the probability of individual propensity score, and match the treatment group and control group according to the method of neighbor matching, to reduce the imbalance between the treatment and control group. As shown in [Fig. 2](#), there is a big difference between the treatment and control group before propensity score matching. In terms of firm ownership, age, size and leverage, the average level of treatment group is significantly lower than the control group. After matching, the imbalance between the treatment and control group declines in different control variables, and the degree of deviation is within 5%. This indicates that the propensity score matching could better reduce the differences between groups, and make the control group more comparable for the treatment group. The results contained in this table mimic that of [Table 2](#), in support of our argument. The significant results imply that the estimated results are valid.

One key criterion for the difference-in-differences approach is that the treatment group and control group conform to the assumption of common trend. As long as this assumption is satisfied, the control group can be treated as a fair reference of the treatment group, in order to ensure a consistent estimation of average treatment effect. To test this and describe the dynamic effects of industry policy on firm innovative activities, [Table 4](#) presents the results of dynamic effects, with robust standard errors adjusted for industry level clustering. As can be observed from columns 1 and 2, the number of granted invention patents between the treatment and control group does not exist a significant difference before the policy was implemented. This indicates that the difference-in-differences model satisfies the assumption of common trend. After the policy implementation, the policy of strategic emerging industries significantly increases the number of granted invention patents, showing a steady trend. This to some extent

¹¹ Stand errors clustered at the industry level are applied for the main identification. However, since the data is firm-level panel data, the standard error clustered at the firm level may be considered following a general practice of DID analysis. Therefore, we have provided robustness checks for the baseline estimation with standard error clustered at the firm level in Appendix Table A1.

Table 5
Change in applying patents.

DEP VAR=log of	(1) applying patent+1	(2) applying invention patent+1	(3) applying utility patent+1	(4) applying design patent+1
Teat × Post	0.177* (0.105)	0.236** (0.095)	0.105 (0.073)	0.037 (0.039)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes
Observations	36,079	36,079	36,079	36,079
R-squared	0.036	0.050	0.028	0.004
Number of firms	4,161	4,161	4,161	4,161

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

Table 6
Heterogeneity — city characteristics.

DEP VAR=log of	(1) granted patent+1	(2) granted invention patent+1	(3) granted utility patent+1	(4) granted design patent+1	(5) granted patent+1	(6) granted invention patent+1	(7) granted utility patent+1	(8) granted design patent+1
Panel A	Higher level of IPR protection				Lower level of IPR protection			
Teat × Post	0.341** (0.140)	0.371*** (0.124)	0.174* (0.093)	0.114** (0.050)	0.166 (0.106)	0.272*** (0.077)	0.086 (0.089)	0.024 (0.044)
Observations	19,735	19,735	19,735	19,735	16,344	16,344	16,344	16,344
R-squared	0.038	0.063	0.024	0.009	0.074	0.097	0.059	0.008
Panel B	Higher level of innovation				Lower level of innovation			
Teat × Post	0.323** (0.162)	0.320** (0.124)	0.168 (0.124)	0.118** (0.057)	0.194** (0.080)	0.278*** (0.060)	0.110 (0.071)	0.036 (0.035)
Observations	19,706	19,706	19,706	19,706	16,373	16,373	16,373	16,373
R-squared	0.030	0.052	0.020	0.008	0.073	0.094	0.058	0.008
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

shows a sustainable effect of the strategic emerging industries policy on firm innovation (invention patents in particular). Moreover, it is advisable to include a parallel trend graph depicting the trend over time for both the treatment group and the control group. This addition will help ascertain whether the parallel trend assumption is met. Fig. 3 depicts the trend and confirms this assumption.¹²

Since different provinces would formulate their five-year plans and implement industry policy regarding specific industries, to motivate corporate innovation. As a result, the estimated results are likely to be affected by local industry policy. In particular in 2010, as the end year of the Eleventh Five-Year Plan, the innovation industry might conduct strategic innovation behavior to achieve their target. For example, they could increase the number of invention patents to increase the number of total patents. Therefore, in column 3 of Table 4 we attempt to control the interaction of province and year fixed effects. The results contained in this table confirm that the increased innovation is solely caused by the adoption of the strategic emerging industries policy.

The empirical analysis presented above uses the number of granted patents as the key dependent variable. This flow variable in time reflects the change in firm innovative output. To fully explore the effect of the strategic emerging industries policy, this paper uses the effective number of patents, or in other words, the number of existing patents, as an alternative dependent variable to analyze the effect of industry policy. Firm existing patents are not only affected by the annual increase in the number of granted patents, but also related to their patent indemnification. In general, if the payment of maintenance fee was overdue, then the patent right will no longer be in force upon the expiration of annual fee. Table 5 contains the results of existing patents. As can be seen, after the consideration of patent maintainment, only the effective number of invention patents rises significantly. While the number of granted utility patents increases, this does not result in a rise in the existing patents of the corresponding category. Our results state that the policy of strategic emerging industries mainly promotes firm existing invention patents, and has a positive impact on the improvement of firm comprehensive power of innovation.

In order to avoid that the estimated results in this paper are generated by contemporary stochastic factors, we follow the methods by Li et al. (2016) and Cantoni et al. (2017) to conduct a placebo test. To be specific, according to our classification, there are 1,212 strategic emerging industries listed firms. Therefore, during the process of random allocation in this paper, we randomly select 1,212 listed firms from the sample (4,161 listed firms) as the treatment group, and use the rest as the control group, to

¹² As for robustness, we have provided parallel trend test with standard error clustered at the firm level in Appendix Figure A1.

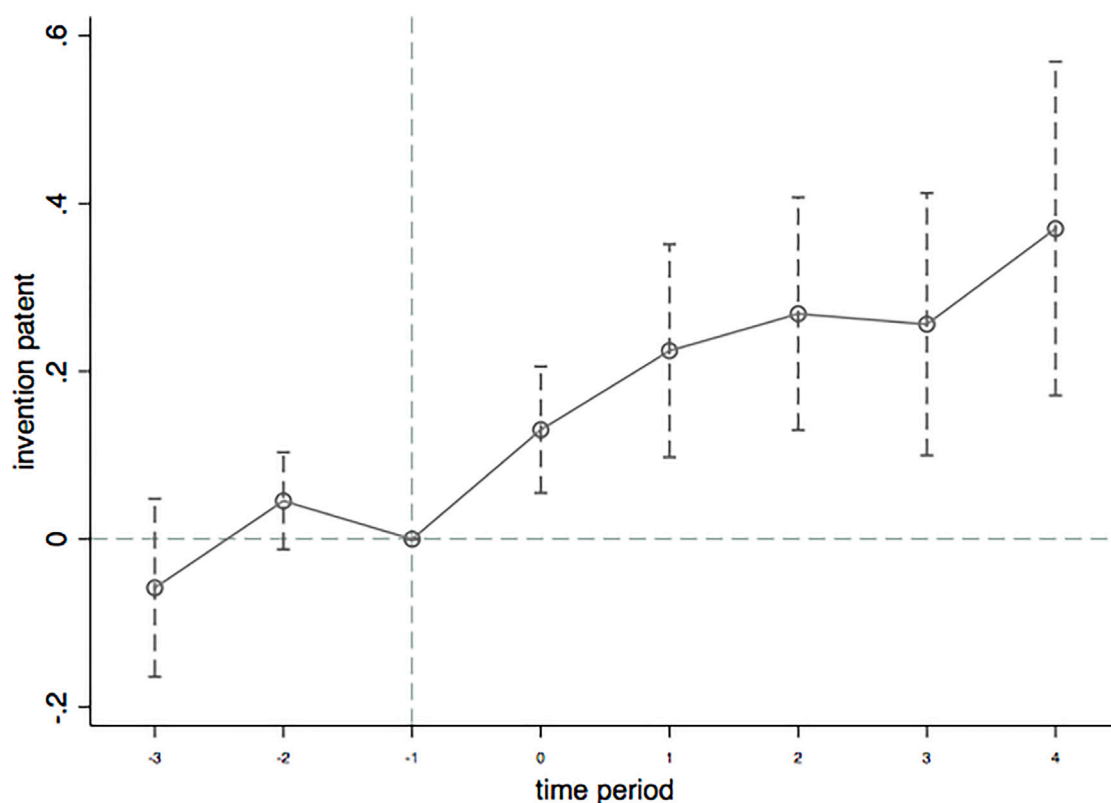


Fig. 3. Parallel trend test.

conduct the difference-in-differences estimation utilizing year 2010 as the policy node. Then we repeat 1,000 times of the fixed effects estimation, and extract the t -statistic in the placebo test. Fig. 4 presents the results. Following La Ferrara et al. (2012), the distribution of the value of the key coefficients and the comparison with the actual value in basic estimation are also reported. As we can see, as for the indicators of firm innovation, especially the number of total granted patents, and the number of granted invention patents, the distribution of t -statistic (of the coefficient we are interested in) is mostly falling into the surrounding of the value of 0, and their t -statistic is mostly less than that in the actual regression (1.90, 2.84) in Table 2. This states that the estimated results are unlikely to be affected by contemporary stochastic factors, and our results are robust.

4.3. Heterogeneity analysis

Table 6 presents the heterogeneity analysis ascribed to city characteristics and use the same specification as Table 2. The analysis investigates the relationship across the characteristics of the firm's location, such as the level of intellectual property rights (IPR) protection, and the level of innovation. As one fundamental institution of the protection of technological innovation outcome, intellectual property system can protect firm innovation achievement via the way of clarification of property rights. This to some extent encourages firms to innovate by means of collaborative innovation and increased innovation input (Ang et al., 2014; Fang et al., 2017). Therefore, Panel A of Table 6 splits the observations based on the indicator of IPR protection, measured by the ratio of city-level IPR infringement cases to national effective patents. As we can see, the industry policy has a positive influence on innovative outcomes by firms located in different levels of IPR protection. In the subsample of higher IPR protection level this positive relationship holds more significantly, and the number of granted invention patents enjoys a higher jump. Besides, if the region where the firm is located has a higher level of innovation, then this firm is more likely to be exposed to new technique and knowledge. The spillover effects of these innovative outcomes can bring about positive impacts on firm innovation. As a result, Panel B separates the sample according to the number of invention patent applications at the city level. Our results confirm that innovation environment plays a crucial role in firm innovative activities.

Table 7 further contains the heterogeneity analysis ascribed to firm characteristics. Different characteristics among firms result in heterogeneous effects of the industry policy. Firm development stage normally indicates the budget constraint. In general firms in later development stage are endowed with relatively sufficient funds, and thus they are less likely to be affected by the cost. In contrast, firms in earlier development stage are relatively constrained by the cost, and endowed with weak institutional standard. As the earlier stage of innovative activities requires large amounts of investment, firm development stage will have an effect on

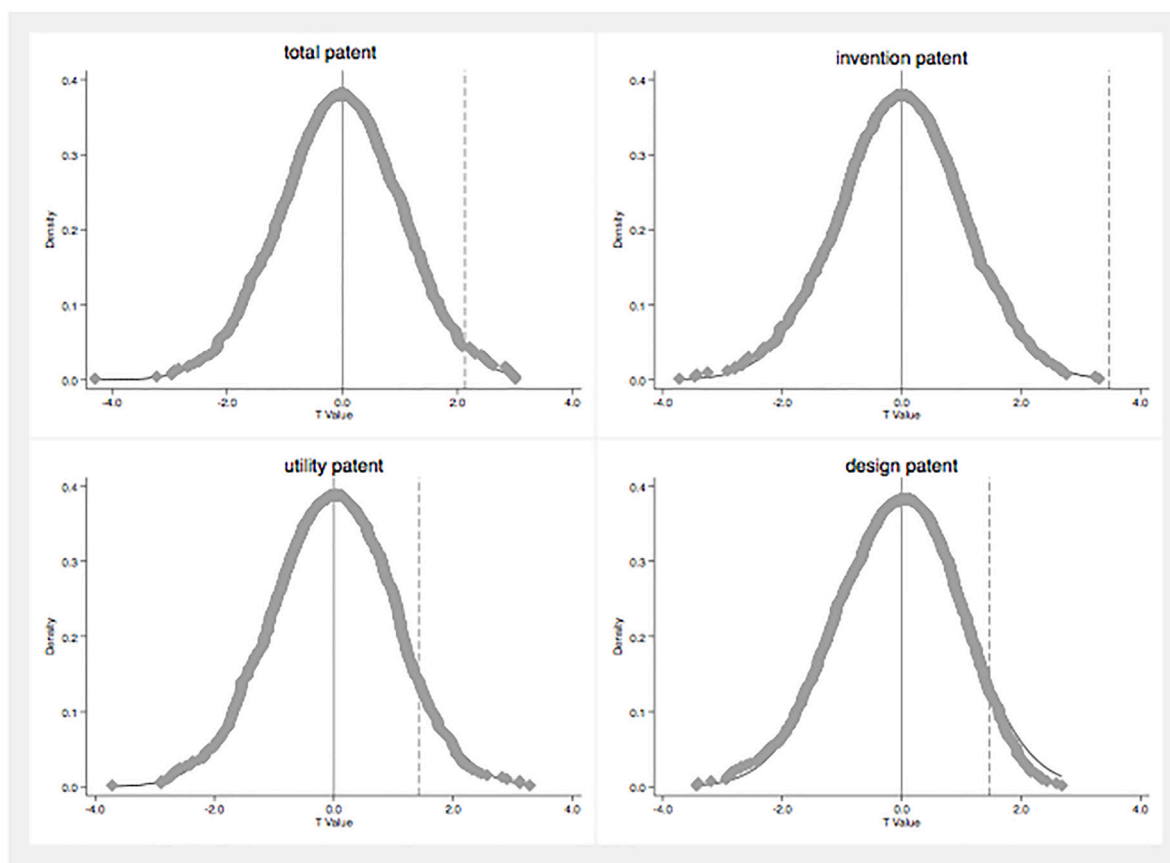


Fig. 4. Placebo test.

Table 7
Heterogeneity — firm characteristics.

DEP VAR=log of	(1) granted patent+1	(2) granted invention patent+1	(3) granted utility patent+1	(4) granted design patent+1	(5) granted patent+1	(6) granted invention patent+1	(7) granted utility patent+1	(8) granted design patent+1
Panel A	Earlier stage				Later stage			
Teat × Post	0.158 (0.128)	0.352*** (0.111)	0.043 (0.104)	0.089 (0.054)	0.260** (0.113)	0.257*** (0.083)	0.147* (0.082)	0.033 (0.042)
Observations	19,497	19,497	19,497	19,497	16,582	16,582	16,582	16,582
R-squared	0.069	0.099	0.047	0.010	0.044	0.069	0.033	0.005
Panel B	SOE				Non-SOE			
Teat × Post	0.255** (0.104)	0.339*** (0.084)	0.120 (0.078)	0.082* (0.045)	0.243 (0.149)	0.299** (0.117)	0.134 (0.123)	0.041 (0.057)
Observations	12,306	12,306	12,306	12,306	23,773	23,773	23,773	23,773
R-squared	0.099	0.128	0.068	0.014	0.033	0.060	0.024	0.007
Panel C	More scientific background executives				Less scientific background executives			
Teat × Post	0.288** (0.129)	0.377*** (0.113)	0.179* (0.104)	0.072 (0.056)	0.197* (0.114)	0.245*** (0.086)	0.066 (0.085)	0.045 (0.038)
Observations	24,703	24,703	24,703	24,703	11,376	11,376	11,376	11,376
R-squared	0.049	0.080	0.034	0.009	0.063	0.093	0.046	0.006
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

Table 8
Mechanism.

DEP VAR=log of	(1) tax refunds+1	(2) total loan+1	(3) Lerner+1
Teat × Post	0.626** (0.294)	0.513** (0.225)	0.014* (0.008)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes
Observations	31,295	27,814	31,401
R-squared	0.182	0.221	0.123
Number of firms	3,899	3,353	4,080

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

Table 9
Mechanism.

	(1) R&D intensity	(2) Graduate ratio	(3) ln(average wage+1)
Teat × Post	0.108*** (0.021)	0.007* (0.004)	0.022 (0.023)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes
Observations	36,079	18,115	33,591
R-squared	0.257	0.313	0.595
Number of firms	4,161	3,071	4,005

Notes: Estimations use panel regression with firm fixed effects and year dummies, and robust standard errors are clustered by industry in parentheses. *, **, and *** respectively denote significance levels at 10%, 5% and 1%.

innovation. When the sample is separated in Panel A, it becomes clear that the industry policy has a structure optimization effect on later stage firms. In other words, this leads to an adjustment of the innovative outcomes structure - a larger increase in the number of granted invention patents in earlier stage firms. There is a rise trend in the number of total granted patents in later stage firms, followed by invention and utility patents.

In addition to firm development stage, its ownership also induces heterogeneous effects in response to the policy of strategic emerging industries. Panel B of Table 7 contains results for firm classified as belonging to state-owned or non-state-owned enterprises (Choi et al., 2011; Fang et al., 2017). The industry policy has a structure optimization effect on non-state-owned enterprises, whilst the number of total granted patents in state-owned enterprises rises accompanied with an adjustment of the innovative outcomes structure. We interpret this as that state-owned enterprises are endorsed by the government and they are also endowed with relatively sufficient funds due to internal soft budget constraint. Besides, if the firm has more executives with scientific background, then this firm is more likely to utilize the industrial policy. As a consequence, Panel C splits the sample according to whether the ratio of executives with scientific background to total executives is more than 10%. Our results confirm that the scientific background of executives shapes firm innovative activities.

4.4. Mechanism analysis

The study highlights the significant role of government support in fostering corporate innovation. By establishing clear standards and policy objectives, the government can offer enterprises opportunities for technological transformation and sustainable development, thereby creating a conducive business environment for innovation activities (Ashford, 2000; Bossink, 2002; Nelson & Rosenberg, 1993; Porter, 1998). Moreover, given that innovation requires substantial funding, involves long investment cycles, and entails high uncertainty in returns, enterprises face considerable financial pressure. Public funds are regarded by enterprises as a supplement to other external financing, and obtaining public funds and financial incentives is effective for improving enterprises' ability to introduce innovation (Cecere et al., 2020; Hall & Lerner, 2010).

Since the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries* clearly put forward to enhance the support of finance and taxation policies in order to promote the innovative development of strategic emerging industries, we assess the impact of the industry policy on innovation through potential mechanisms in Table 8. The first channel is through which firms receive tax refunds from the government. Firms with lower tax costs have an incentive to conduct research and development activities. From the point of view of expansion decision, we explore whether and how the industry policy affects the level of loan of firms. Here we promote loan as a measure of credit support, as firms need to finance funds to expand production. The last channel we examine is market competition. The Lerner index is used to measure market competition. We aim to test whether the industry policy raises the industrial concentration, or enhances the monopoly position of top firms. Throughout the table, we find evidence that the potential mechanism that the industry policy affects innovative activities is by means of an increase in tax support, a relaxation of financing restriction and market competition, in support of the argument proposed.

The analysis above identifies tax benefits and loans as key mechanisms driving innovation. However, it lacks a comprehensive analysis of how these mechanisms interact or affect different types of firms. For instance, large firms and smaller firms may experience varying benefits from loans, while state-owned enterprises may respond differently to tax incentives compared to private firms. A deeper exploration of these distinctions would offer a more nuanced understanding of the policy's impact on innovation across different firm types. Therefore, Appendix tables A2 and A3 respectively show the differences in the support provided by different industrial policy tools to enterprises of different ownerships and scales. As can be seen, state-owned enterprises are most affected by credit support, while non-state-owned enterprises receive the greatest support from tax incentives. This may be because state-owned enterprises have national credit endorsements and therefore face soft budget constraints, making it relatively easier to obtain credit support. In addition, smaller enterprises are more affected by credit support, while larger enterprises are more affected by tax refunds. This may be because smaller enterprises face tighter budget soft constraints and therefore receive greater credit support after the implementation of industrial policies, while larger enterprises mainly face the problem of converting more revenue into profits and are therefore more affected by tax incentives after the implementation of policies.

For the mechanism of market competition measured by the Lerner index, the market competition may serve as a mediation but not a channel because the market structure may need time to change. With different market competition, firms may get different performances after implementing the policy. There, we analyze the degree of industry competition as the moderating effect of industrial policies in Appendix Table A4. The results indicate that in industries with low Lerner index, that is, industries with sufficient competition, industrial policies can significantly improve the innovation level of enterprises, especially the growth rate of invention patents is the fastest. In high Lerner index industries, which have a high degree of monopoly, although industrial policies can also promote the improvement of enterprise innovation level, the growth rate of invention patents is weaker than that of low Lerner index industries, indicating that creating a moderate competitive environment has an important impact on the effectiveness of industrial policies.

In addition, tax refunds and loans may be channels for many outputs, such as productivity and sales, not just innovation. It would be better if we could find channels more directly related to innovation input, such as the input of R&D, the quality of the innovator, etc. Therefore, we include relevant analysis on the impact of strategic emerging industry policies on enterprise R&D investment and human capital composition in Table 9. From the estimation results in columns 1 and 2, it can be seen that the policy of strategic emerging industries has promoted the increase of R&D investment in enterprises, significantly improving the intensity of R&D investment and improving the human capital structure of enterprises, resulting in a significant increase in the proportion of employees with postgraduate degrees or above to the total number of employees. In addition, when considering employee compensation as a measure of labor productivity, the estimated results in column 3 indicate that strategic emerging industry policies have also promoted the improvement of labor productivity in enterprises, but this result lacks statistical significance. Throughout the table, we find evidence that the potential mechanism that the industry policy affects innovative activities is by means of an increase in R&D input and the quality of the innovator.

5. Conclusion and policy implications

Utilizing the data of listed companies in China A-shares market, this paper adopts a difference-in-differences approach to empirically examine how the policy of strategic emerging industries affects firms' innovative outcomes, in the years around the prevalence of the *State Council's Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries*.

We find three main findings. First, the increase in innovation is significantly larger in firms belonging to the strategic emerging industries than that without this title. On average, the number of total granted patents rises by 27.8% in the strategic emerging industries compared to the counterpart. Second, we find evidence that the industry policy significantly promotes tax benefits, firm loans, industrial concentration, R&D input and innovator quality, leading to more innovative outcomes. Third, the rise in innovation is greater in firms located in a city with higher levels of IPR protection, and innovation. As for firm characteristics, the industry policy has a structure optimization effect on earlier stage or non-state-owned enterprises, whilst the number of total granted patents in later stage or state-owned enterprises rises accompanied with an adjustment of the innovative outcomes structure. The scientific background of executives also plays a crucial role in firm innovative activities.

Our results contribute to the literature in two ways. First, we provide evidence that the strategic emerging industries policy promotes firm innovative outcomes in China. The strategic emerging industries policy plays a key role in China's economic transition, given that the price of production factors increases in China in an era of economic transition. Second, we show this industry policy as one new explanation of the innovation miracle in China. In addition, this industry policy has a structure optimization effect, which also sheds light on developing countries that aim to enhance their innovation quality. China still lags behind developed countries in terms of innovation. This strategic emerging industries policy has achieved significant results in China and has important reference value for other developing countries. Many developing countries also face the problem of difficult independent innovation, such as India and Vietnam. Due to insufficient policy support, firms find it difficult to transform from imitation to innovation in these countries. These countries can learn from China's experience, intensify technological innovation and improve the core competitive strength of industry. In addition, Chinese high-tech companies have also begun to expand internationally, collaborating with other developing countries to export successful experiences, helping their local firms enhance their innovative capabilities. China's experience has been replicated and promoted, providing more possibilities for solving the problem of independent innovation for firms in developing countries. Overall, China's strategic emerging industries policy not only promotes high-quality domestic economic development, but also provides useful reference for the world, especially developing countries.

The analysis presented in this paper establishes a robust positive statistical association between the strategic emerging industries policy and firm granted patents, in particular in granted invention patents. This indicates that the policy of strategic emerging industries to some extent stimulates corporate innovative activities. Despite this, as microcosmic bodies, the aim of firms to conduct innovative activities is to make profit. Therefore, this paper still points out further research directions to use the interaction model to examine the impact of different patents on firm value, profitability and total factor productivity after the implementation of strategic emerging industries policy. This reflects the change in firm innovation performance after the industry policy implemented.

CRedit authorship contribution statement

Xiaoming Zhang: Methodology, Software, Formal analysis, Investigation. **Weijie Luo:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – original draft, Resources. **Di Xiang:** Data curation, Visualization.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.iref.2025.103858>.

Data availability

Data will be made available on request.

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